Dr. Dave Eachus
Middle School STEM
Mary, Star of the Sea School
Innovations Lab (C-17)
Email: deachus@starofthesea.org
Phone: (808) 228-2285 x 2221
MSOS 2026-2027

STEM — Grade 6 Syllabus

Grade 6: Earth & Space Science · SY 2026-2027

Mary, Star of the Sea School Mission

Nurture, Unite, Inspire. We are committed to nurturing each child's curiosity, uniting our classroom in respect and collaboration, and inspiring every student to grow in knowledge, faith, and wonder for the world around them.

2026-2027 Essential Question

"How can I model kindness, fairness, and humility like Jesus?"

Course Description

6th Grade STEM is an introduction to Earth and Space Science. It is designed to unite our world view as citizens of the planet, nurture our understanding of how the Earth's systems are interrelated and inspire the next generation of scientists and engineers.

The year begins with a short introduction to the Scientific Method and the Engineering Design Process as well as the principal tools and techniques of a scientist and engineer. Then it's time to leave this planet we call home and explore the wonders of the Universe. We will return to the "Big Bang" and discover the universe's beginnings. After the Winter Break, we will be going back in deep time, 4.5 billion years ago to discover how the Earth was made and learn the key events that shaped our world. After Spring Break, we will conclude the year with an investigation of two pressing global issues, climate change and the low-energy future.

Teacher

Dr. Dave Eachus

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Classroom & Lab Materials

- Composition notebook (lab journal)
- Pencils, pens, colored pencils
- Lab safety goggles
- Computer / device for simulations
- Calculator

Scope & Sequence — Grade 6: Earth & Space Science

2026–2027, August–June. All unit dates are projected and may be subject to change.

Unit 1 — Why STEM   3 Aug – 28 Aug

Key Concept: Systems · Related: Patterns, Form, Function · Global Context: Science and Technical Innovation

Inquiry statement: The Scientific Method and the Engineering Design Process allow us to investigate the form and function of patterns observed in natural systems or designs.

Factual: How do scientists and engineers use the Scientific Method and the Engineering Design Process?

Conceptual: How does experimental science and engineering expand our body of knowledge?

Debatable: Can the Scientific Method or the Engineering Design Process be used to answer any question or solve any problem?

Know: Scientific Method, Engineering Design Process, Extinction Level Events

Understand: How the Scientific Method is used to answer questions. The linkage between ELE’s and mass extinctions.

Do: Lab Safety, Scientific Method/Engineering Design Process, variables and hypothesis formulation, Lab report format.

Summative Assessment Tasks:

- ELE Poster (Crit. A/D)

Unit 2 — Our Universe   31 Aug – 18 Dec

Key Concept: Change · Related: Models, Interaction, Movement · Global Context: Science and Technical Innovation

Inquiry statement: Models allow scientists to understand the movements and interactions that change our Universe.

Factual: What forces govern our Universe?

Conceptual: How was our Universe created?

Debatable: How old is our Universe?

Know: Structure and states of matter, types of objects in the universe, how we use models to help us understand our Universe.

Understand: How the Universe was formed and that it is still expanding, formation of stars, importance of gravity, scale of the Universe.

Do: Mass Lab, Volume Lab, Density Lab, Alien in Me, Great Pluto Debate.

Summative Assessment Tasks:

- Density Lab (Crit. B/C)
- Great Pluto Debate (Crit. A/D)

Unit 3 — How the Earth was Made   7 Jan – 12 Mar

Key Concept: Systems · Related: Evidence, Environment · Global Context: Orientation in Space and Time

Inquiry statement: Evidence allows us to see that the interplay of geologic, environmental, and biological systems created our Earth.

Factual: Where is the evidence of the 4.5 billion year development of the Earth?

Conceptual: How do we know the age of our Earth?

Debatable: What are the key events that resulted in our world being what it is today?

Know: 25 key events in the 4.5 billion year history of our Earth.

Understand: How the key events shaped our Earth into what we see today.

Do: Explore the evidence in the rock, investigate how our Earth changed over the past 4.5 billion years, and understand what fossils tell us about life in the deep past. Rock Lab, Fossil Lab.

Summative Assessment Tasks:

- Ages of the Earth Presentation (Crit. A/D)
- The Great Oxygenation Lab (Crit. B/C)
- Earth Timeline Project (Crit. A/D)

Unit 4 — Climate Change and the Low-Energy Future   29 Mar – 3 Jun

Key Concept: Change · Related: Balance, Energy, Consequences · Global Context: Globalization and Sustainability

Inquiry statement: Exploring mankind’s use of energy and its link to our climate will reveal a delicate balance of forces that are changing and will have significant consequences for the sustainability of life on our planet.

Factual: What factors affect climate?

Conceptual: How is mankind's use of energy changing climate factors? How will a low-energy future affect our ability to deal with climate change?

Debatable: Is human activity having a significant impact on climate change?

Know: The evidence behind Climate Change and the Low Energy future.

Understand: How the factors affecting climate change are linked to energy usage.

Do: Identify the difference between weather and climate, explore the factors that are causing Earth's climate to change, investigate the impact of those changes on our Earth, identify where the energy comes from and how it is used, explore trends in energy use growth and its implications for the future, investigate the nexus created by the changing climate and the potential to run out of energy in the near future.

Summative Assessment Tasks:

- Critical Lab Skills Test (Crit. B/C)
- Sources of Energy Presentation (Crit. A/D)
- Baseball Project (Crit. A/D)

STEM Assessment Criteria

Each criterion is scored on an MYP-style scale of 0–8. Student activities are assessed against the criteria below; end-of-quarter letter grades reflect overall progress.

Criterion	Focus	Description	Scale
A	Knowing & Understanding	Recall and apply scientific knowledge; explain phenomena.	0–8
B	Inquiring & Designing	1st half of the Experimental / Design Cycle: question, plan, hypothesize, design.	0–8
C	Processing & Evaluating	2nd half of the Experimental / Design Cycle: collect, analyze, draw conclusions.	0–8
D	Reflecting on the Impacts of Science / Engineering	Consider ethical, environmental, and social implications of science and engineering decisions.	0–8

Learning Support & Communication

Parents with questions or concerns may directly contact the instructor.

Students are encouraged to use Flex Time or lunch recess to obtain additional learning support.

The Innovations Lab is open every Wednesday afternoon (except Mini-days) until 4 pm for any student needing help or to finish lab work.

Students and/or parents should email Dr. Eachus to make arrangements for additional learning opportunities.

Empowered to Succeed, Guided by Faith